

```
In [3]: #Datastructure
```

```
In [ ]: # List Datastructrure in Python
```

```
In [4]: l=[]  
print(l)
```

```
[]
```

```
In [5]: type(l)
```

```
Out[5]: list
```

```
In [6]: print(type(l))
```

```
<class 'list'>
```

```
In [16]: l.append(10)  
l
```

```
Out[16]: [10]
```

```
In [17]: l.append(20)  
l.append(30)  
l.append(40)  
l.append(50)  
l
```

```
Out[17]: [10, 20, 30, 40, 50]
```

```
In [14]: len(l)
```

```
Out[14]: 5
```

```
In [15]: l.clear()
```

```
In [19]: l1=[10,'2.3',1+2j,True,'nit',[1,2,3]]
```

```
In [20]: l1
```

```
Out[20]: [10, '2.3', (1+2j), True, 'nit', [1, 2, 3]]
```

```
In [21]: len(l1)
```

```
Out[21]: 6
```

```
In [22]: print(l)  
print(l1)
```

```
[10, 20, 30, 40, 50]
```

```
[10, '2.3', (1+2j), True, 'nit', [1, 2, 3]]
```

```
In [25]: print(id(l))  
print(id(l1))
```

```
2425017249920
```

```
2425020190016
```

```
In [26]: l2= l.copy()  
12
```

```
Out[26]: [10, 20, 30, 40, 50]
```

```
In [27]: print(id(l))  
print(id(l1))  
print(id(l2))
```

```
2425017249920  
2425020190016  
2425033037120
```

```
In [28]: len(l2)
```

```
Out[28]: 5
```

```
In [29]: l2.clear()
```

```
In [30]: len(l2)
```

```
Out[30]: 0
```

```
In [31]: del l2
```

```
In [32]: len(l2)
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[32], line 1  
----> 1 len(l2)  
  
NameError: name 'l2' is not defined
```

```
In [33]: l.count(20)
```

```
Out[33]: 1
```

```
In [34]: l.count(20)
```

```
Out[34]: 1
```

```
In [35]: l.count(100)
```

```
Out[35]: 0
```

```
In [36]: a =[1,2,3]  
b=[4,5,6]  
a.extend(b)  
print(a)
```

```
[1, 2, 3, 4, 5, 6]
```

```
In [38]: l2 = []  
12
```

```
Out[38]: []
```

```
In [39]: l2.extend(l)
```

```
In [40]: l2
```

```
Out[40]: [10, 20, 30, 40, 50]
```

```
In [41]: print(l)
         print(l1)
         print(l2)
```

```
[10, 20, 30, 40, 50]
[10, '2.3', (1+2j), True, 'nit', [1, 2, 3]]
[10, 20, 30, 40, 50]
```

```
In [42]: l[0]
```

```
Out[42]: 10
```

```
In [43]: l[0]=100
```

```
In [44]: l
```

```
Out[44]: [100, 20, 30, 40, 50]
```

```
In [46]: job=[5000]
         job[0]
         job[0]=10000
         job
```

```
Out[46]: [10000]
```

```
In [47]: l[4]=500
         l
```

```
Out[47]: [100, 20, 30, 40, 500]
```

```
In [48]: l.index(100)
```

```
Out[48]: 0
```

```
In [50]: l.append(100)
```

```
In [51]: l
```

```
Out[51]: [100, 20, 30, 40, 500, 100, 100]
```

```
In [52]: l.index(100) # Even multiple 100 value present in the list still it print the fi
```

```
Out[52]: 0
```

```
In [ ]: l.insrt()
```

```
In [53]: l.insert(2,25)
         l
```

```
Out[53]: [100, 20, 25, 30, 40, 500, 100, 100]
```

```
In [54]: 12
```

```
Out[54]: [10, 20, 30, 40, 50]
```

```
In [55]: 12.pop()
```

```
Out[55]: 50
```

```
In [56]: 12
```

```
Out[56]: [10, 20, 30, 40]
```

```
In [58]: 12.pop(2)
```

```
Out[58]: 30
```

```
In [60]: 12
```

```
Out[60]: [10, 20, 40]
```

```
In [61]: 12.remove(40)
```

```
In [62]: 12
```

```
Out[62]: [10, 20]
```

```
In [63]: 12.reverse()
```

```
In [64]: 12
```

```
Out[64]: [20, 10]
```

```
In [65]: 12.sort()
```

```
In [66]: 12
```

```
Out[66]: [10, 20]
```

```
In [67]: 12.sort(reverse =True)  
12
```

```
Out[67]: [20, 10]
```

```
In [68]: 12.sort(reverse =False)  
12
```

```
Out[68]: [10, 20]
```

```
In [69]: 13=[10,20,[2,3],3]  
13[2]=[2]  
13
```

```
Out[69]: [10, 20, [2], 3]
```

```
In [71]: l3=[10,20,[2,3,4,5],3]
         l3[2]=[2,4,5]
         l3
```

```
Out[71]: [10, 20, [2, 4, 5], 3]
```

```
In [72]: l4=[10,20,30,40,50,500,100]
         10 in l4
```

```
Out[72]: True
```

```
In [74]: for i in l4:
         print(i)
```

```
10
20
30
40
50
500
100
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```